BE (E&TC)	PLC SCADA and Automation LAB	
Experiment No: 6	Design simulation of cylinder piston pump using pneumatic kit using Virtual lab	**Pages**

AIM: Design simulation of cylinder piston pump using pneumatic kit using Virtual lab

OBJECTIVE:

To design and implement the control of a pneumatic system using a PLC for industrial automation applications.

THEORY:

Pneumatics is successfully used in many areas of industrial automation. **Pneumatic Trainer** facilitates the students and the industrial professionals to explore the fundamental of "pneumatics". It tells how one device can be used to manage, command, direct or regulate the behaviour of other system.

Pneumatics is a branch of science that is related with *Air*. In pneumatics the working medium is "Compressed Air". Now - a - days this working medium is becoming very popular in industries like chemical, petroleum, gases, food & beverage, robotics, machineries etc. The reason for being so popular is that it is fire free and non-hazardous to our environment. And it is abundantly available in nature. It is very easy, simple and safe to operate. Pneumatic Trainer assist user to understand the principle performance of the pneumatic components as well as different actuators. Another additional feature of Pneumatic Trainer, it has both manual and automatic based control

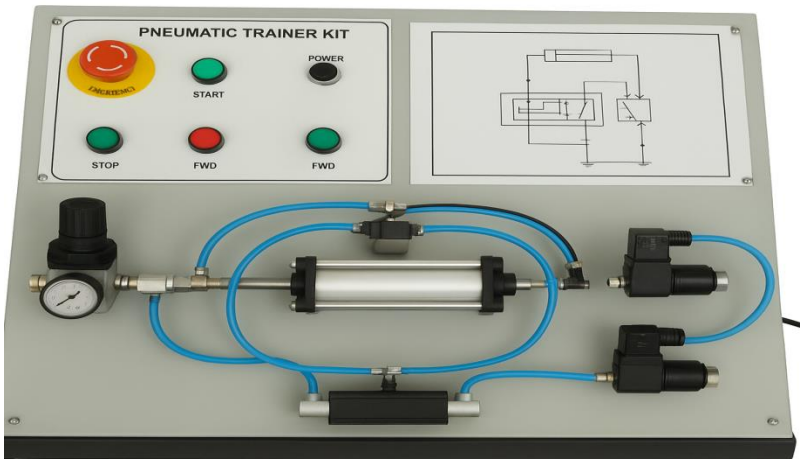


Fig.1- Pneumatic Trainer Kit

Pneumatic Trainer Kits are known for their accuracy and precision. The Pneumatic Trainer Kits are well tested on the stringent industry parameters by the team of experts which assures the quality of the final product. Pneumatic systems are used in controlling train doors, automatic production lines, mechanical clamps, etc Fig. 1.

Pneumatic Trainer provides different types of valves such as electronic and mechanical. This helps to understand the user about manual and automatic process. There are wide range of experiments provided on the trainer that can be performed.

Compressor: -

A compressor can compress air to the required pressures. It can convert the mechanical energy from motors and engines into the potential energy in compressed air (Fig. 2). A single central compressor can supply various pneumatic components with compressed air, which is transported through pipes from the cylinder to the pneumatic components. Compressors can be divided into two classes: reciprocator and rotary.



Fig.2- Compressor

The reason for using pneumatics, or any other type of energy Transmission on a machine, is to perform work. The accomplishment of work requires the application of kinetic energy to a resisting object resulting in the object moving through a distance. In a pneumatic system, energy is stored in a potential state under the form of compressed air. Working energy (kinetic energy and pressure) results in a pneumatic system when the compressed air is allowed to expand.

Link of virtual lab:

https://share.google/UXCoBuUNGnaUEiNPz

Procedure to run the simulator

1. Enter the value of load you want to transfer in LOAD tab. \
2. Enter the value of angle of inclination on which object is kept in ANGLE tab. \
3. Enter speed of extension of cylinder at witch you want to transfer the object in SPEED tab. \
4. Using formula calculates the cylinder bore diameter and enter the answer in the DAIMETER tab. \
5. Submit the answer. \
6. If your calculation is wrong, you will be asked to repeat the same. Refer FORMULA tab and verify your calculations. \
7. Select the direction control valve. \

8. Complete connection with working port, supply port and exhaust port. \
9. Run the system.

CONCLUSION:

QUESTIONS:

1. What is the main advantage of hydro pneumatics?
2. Name the different pneumatic position sensors.
3. Name the common methods used for designing logic circuits?
4. What is the advantage of using microelectronic control for fluid power compared to electro-mechanical control?

REFERENCES:

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ISO 1219-1:1991 Fluid power systems and components- Graphic symbols and circuits diagrams- Part 1
ISO 3320:2013 Fluid Power Systems and Components- Cylinder bores and piston rod diameter and area ratios
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